

Practical No. 7

SharedPreferences – Simple Login App

Aim

To develop a simple Android application using Kotlin to create a login page, save username using SharedPreferences, and display “Welcome back <username>” when the app reopens.

Software Requirements

- Android Studio
 - Kotlin
 - Android SDK
-

Step 1: Create a New Android Studio Project

- Language: Kotlin
 - Template: Empty Views Activity
 - Min SDK: API 21 (or default)
 - File Name: loginapp
-



Step 2: Layout (activity_main.xml)

Very basic UI:

- One EditText (Username)
- One EditText (Password)
- One Button (Login)
- One TextView (Welcome Message)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <EditText
```

```

        android:id="@+id/etUsername"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Username" />

<EditText
    android:id="@+id/etPassword"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter Password"
    android:inputType="textPassword" />

<Button
    android:id="@+id/btnLogin"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Login" />

<TextView
    android:id="@+id/tvWelcome"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:textSize="18sp"
    android:paddingTop="20dp" />

</LinearLayout>

```

Step 3: Kotlin Code (MainActivity.kt)

This is the entire logic. Simple and direct.

```

package com.example.loginapp

import android.content.Context
import android.os.Bundle
import android.widget.*
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    lateinit var etUsername: EditText
    lateinit var etPassword: EditText
    lateinit var btnLogin: Button
    lateinit var tvWelcome: TextView

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        etUsername = findViewById(R.id.etUsername)
        etPassword = findViewById(R.id.etPassword)
        btnLogin = findViewById(R.id.btnLogin)
        tvWelcome = findViewById(R.id.tvWelcome)

        val sharedPreferences = getSharedPreferences("MyPref",
            Context.MODE_PRIVATE)

        // Check saved username when app opens
    }
}

```

```

        val savedUsername = sharedPref.getString("username", "")

        if (!savedUsername.isNullOrEmpty()) {
            tvWelcome.text = "Welcome back $savedUsername"
        }

        btnLogin.setOnClickListener {

            val username = etUsername.text.toString()

            val editor = sharedPref.edit()
            editor.putString("username", username)
            editor.apply()

            tvWelcome.text = "Welcome back $username"
        }
    }
}

```

Output

- User enters username and password
- Click Login
- Username is saved using SharedPreferences
- Close the app
- Reopen the app
- App displays:

👉 **Welcome back <username>**

